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Project Final Report

1. Introduction

1.1 Research Background

Cardiovascular disease, as a major global health threat, presents significant challenges to medical diagnosis due to its complexity and diversity. According to the 2019 Global Health Assessment Report released by the World Health Organization, heart disease accounted for 16% of global deaths over the past 20 years, highlighting the severity and prevalence of heart health issues. Heart disease can be specifically divided into multiple subtypes, including coronary artery disease, arrhythmia, heart failure, valvular heart disease, cardiomyopathy, congenital heart disease, pericardial disease, and infectious heart disease. Each type has its own unique pathological mechanisms and clinical manifestations. Early detection and accurate diagnosis are of crucial importance for effective management and treatment of these diseases, especially in reducing the mortality rate of heart disease patients.[1]

Since its invention by Einthoven in the early 20th century, the electrocardiogram (commonly abbreviated as ECG) has become an important tool for evaluating heart function and diagnosing heart disease[2,3]. Electrocardiogram technology records the changes in electrical activity generated by the heart during each cardiac cycle on the body surface through an electrocardiograph. After more than 100 years of development, this technology has become increasingly mature. The mainstream technology currently used in clinical practice is the 12-lead electrocardiogram system, which consists of 6 limb leads (I, II, III, aVR, aVL, aVF) and 6 chest leads (V1-V6), and can accurately display cardiac activity. The 12-lead electrocardiogram provides a comprehensive view of cardiac electrical activity in three-dimensional space, with each lead capturing electrical activity information of specific areas of the heart from different angles, collectively forming a comprehensive evaluation system for cardiac function.

However, although electrocardiogram images can accurately reflect the electrophysiological state of the heart, interpreting a 12-lead electrocardiogram requires a lot of professional knowledge and clinical experience. [4,5] The interpretation of electrocardiogram not only requires a deep understanding of normal electrocardiogram patterns, but also the ability to identify various abnormal patterns and their clinical significance. [6] This professional requirement poses a huge challenge in areas where there is a lack of electrocardiography experts or limited diagnostic capabilities in primary healthcare institutions and emergency departments. In these environments where medical resources are relatively scarce, accurate interpretation of electrocardiograms is often limited, resulting in many heart diseases not being diagnosed and treated in a timely manner.

Therefore, in order to improve the efficiency and accuracy of electrocardiogram diagnosis, automated electrocardiogram analysis technology has become increasingly necessary. [7,8] Especially in the context of an aging population and increasing burden of cardiovascular disease, the development of efficient and accurate automatic

electrocardiogram analysis systems has important practical significance. This type of system can not only assist clinical doctors in making diagnostic decisions, but also provide professional level electrocardiogram interpretation services in remote areas, thereby narrowing the regional gap in medical resources and promoting equalization of medical services.

1.2 Research Objectives

Previously, some research teams have attempted and made breakthroughs in the field of electrocardiogram analysis using artificial intelligence methods. [9-13] For example, Maji [14] proposed a technique for automatically detecting atrial fibrillation (AF), utilizing high-order statistical moments in the empirical mode decomposition (EMD) domain of electrocardiogram (ECG) signals, ultimately achieving a sensitivity of 96%. Venkatesan [15] used an SVM classifier for arrhythmia heartbeat classification, successfully increasing the maximum accuracy between normal individuals and individuals at risk of arrhythmia to 96%. Deep learning technology has also been applied in this field; Ribeiro et al. [16] used Convolutional Neural Networks (CNN) to classify electrocardiographic abnormalities and demonstrated the feasibility of using CNN to classify six different types of electrocardiographic abnormalities.

However, these technologies still have significant limitations. Maji's technology has drawbacks such as lengthy diagnostic processes, high costs, and reliance on manual intervention. Venkatesan's model has high sensitivity and good accuracy in identifying specific arrhythmias, but performs poorly in multi class classification and has limited generalization ability. Ribeiro et al.'s CNN faces limitations such as limited number of misclassifiable anomalies and insufficient robustness across different datasets. In addition, existing methods exhibit significant performance degradation when dealing with complex or atypical cases, which is a serious issue in clinical practice as real-world electrocardiogram data often contains various noises and variations.

To address these limitations, the goal of this study is to improve the existing CNN model of Ribeiro by increasing the number of misclassifiable anomalies and enhancing the performance and robustness of the model. The improved CNN model should be able to improve the accuracy and efficiency of automatic electrocardiogram analysis, especially in areas where there is a shortage of electrocardiogram experts and limited diagnostic capabilities in primary healthcare and emergency departments. Specifically, this study aims to develop a comprehensive system that can simultaneously handle normal/abnormal classification and specific abnormal type recognition, optimizing overall performance through a cascaded classification strategy.

This study adopts the Transformer architecture as the core model, based on its significant advantages in processing sequential data. Compared with traditional CNN models, the self-attention mechanism of Transformer can better capture long-range dependencies in electrocardiogram signals, which is particularly important for identifying rhythm abnormalities involving multiple cardiac cycles. In addition, the scalability of Transformer models enables them to adapt to classification tasks of different complexities, from simple binary classification to complex multi

classification problems.

In summary, using deep learning algorithms to automatically interpret 12 lead electrocardiograms may fundamentally change electrocardiogram diagnosis by improving diagnostic accuracy and workflow efficiency. Based on previous research, this study expanded the number of misclassifiable anomalies and improved the performance and robustness of the DNN model. Through systematic model design and optimization strategies, this study aims to promote the development of electrocardiogram automatic analysis technology towards more accurate and reliable directions.

1.3 Expected outcomes

The above is an introduction to the overall research content and achievement goals of the study. In this study, my task is to use the Transformer model to classify data and adjust the model based on the results to enhance its accuracy and robustness in identifying and diagnosing electrocardiogram results. The expected final result should be able to achieve more accurate judgment of normal and abnormal results, that is, to achieve binary classification of these two types of samples. On this basis, disease diagnosis of abnormal samples is achieved, that is, multi classification of abnormal samples is carried out.

Specifically, this study aims to achieve the following quantitative indicators: in binary classification tasks, the target accuracy is not less than 90%, and the F1 score is above 0.88; In multi classification tasks, the average classification accuracy for 12 major cardiac abnormalities is not less than 85%, and the weighted F1 score is above 0.82. These indicators will be validated through comparative experiments with existing benchmark methods to ensure that the method proposed in this study has a competitive advantage in performance.

In addition to quantitative indicators, this study also expects significant improvements in model robustness. By introducing advanced data preprocessing techniques and regularization methods, the model should demonstrate good adaptability to electrocardiogram data from different sources and with different qualities. Especially when dealing with electrocardiogram signals containing noise, baseline drift, or other common artifacts, the model should maintain stable performance, which is crucial for practical applications in clinical environments.

Another important expected outcome is an improvement in the interpretability of the model. By utilizing the attention mechanism of Transformer architecture, this study will explore methods for visualizing the decision-making process of the model, identifying the electrocardiogram regions and time periods that contribute the most to the classification results. This interpretability analysis not only helps validate the reliability of the model, but also provides valuable diagnostic references for clinical doctors, enhancing their trust in AI assisted diagnostic systems.

From the perspective of technological contribution, this study aims to make progress

in the following areas: firstly, to validate the effectiveness of Transformer architecture in electrocardiogram classification tasks, particularly in terms of its advantages in processing 12 lead electrocardiogram signals; Secondly, specific preprocessing and feature extraction methods suitable for electrocardiogram data are proposed to optimize the utilization efficiency of complex FFT features; Finally, develop an efficient cascade classification strategy that balances computational complexity and classification accuracy, providing feasible solutions for deployment in resource constrained environments.

Ultimately, this study aims to develop a complete prototype of an automated electrocardiogram analysis system that can receive raw electrocardiogram data and output comprehensive diagnostic reports including normal/abnormal judgments and specific types of abnormalities. This system should have a good user interface and integration capability, laying the foundation for subsequent clinical validation and practical deployment.

2. Related Work

2.1 Overview of Existing Solutions

At present, there are mainly two methods used in the field of automated analysis of electrocardiograms, namely statistical based methods and artificial intelligence based methods. Although statistical analysis techniques have high sensitivity, the diagnostic process is lengthy, costly, and still relies on manual intervention; The analysis techniques based on artificial intelligence have limitations in multi classification tasks, and the robustness of the dataset is poor. Next, I will provide a detailed introduction to each of them.

Firstly, the research is based on statistical methods, taking Maji's work [14] in 2013 as a reference. The core of Maji's research method is manual design and feature extraction. The researchers first performed empirical mode decomposition on the denoised single lead ECG signal, decomposing it into a series of intrinsic mode functions. They observed significant differences in the morphology of specific IMFs (such as IMF3, IMF4) between normal heart rhythm and atrial fibrillation heart rhythm, particularly their corresponding P-wave oscillation characteristics. Subsequently, they extracted higher-order statistical features (such as variance and standard deviation) from the selected IMF to quantify this morphological difference. Finally, use a simple linear decision function (such as calculating the perpendicular bisector of two types of center points in the feature space as the decision boundary) to classify the extracted features to distinguish between normal heart rhythm and atrial fibrillation.

This method has relatively lower computational complexity compared to deep learning methods, requires limited computing resources, and has lower equipment requirements, making it more practical to promote technology and equipment in underdeveloped areas. The entire process is relatively transparent, including EMD decomposition and feature extraction, with strong interpretability because the physical meaning of the features (reflecting the presence of P-waves) is relatively clear. And the sensitivity is high, reaching 96%. However, the performance of the entire system heavily relies on

the quality and specificity of manual feature engineering, and the feature selection process may introduce subjective bias, making it difficult to capture complex and deep electrocardiogram patterns. Therefore, relying more on manual intervention to achieve better training results requires high demands on human resources. In addition, the classifier used in this method is very simple, and the model capacity is limited, making it difficult to handle multi class or complex abnormal situations. The generalization ability of the entire method is usually poor, heavily relying on specific datasets (such as MIT-BIH), which may lead to performance degradation on broader and noisy clinical data.

Another approach is an end-to-end learning method based on modern deep neural networks. This method is represented by the Ribeiro team's research in 2020[16]. Their core idea is to apply deep learning techniques to image classification for diagnosis, adopting an end-to-end deep learning paradigm. This model takes raw, unprocessed 12 lead ECG signals as direct input, and automatically learns and extracts multi-level features related to diagnosis through a deep one-dimensional residual neural network, ultimately completing the classification task. This study trained a model on a large-scale dataset of over 2.3 million 12 lead electrocardiograms to diagnose six representative heart rhythm abnormalities and conduction disorders: first-degree atrioventricular block, right bundle branch block, left bundle branch block, sinus bradycardia, atrial fibrillation, and sinus tachycardia.

This research method avoids tedious and potentially incomplete manual feature engineering through end-to-end learning. This model can automatically discover the most effective feature representations from massive data and typically achieves higher performance than traditional methods (F1 scores in this study were all above 80%, with specificity exceeding 99%). Large scale, multi center datasets enable models to have strong generalization ability and are closer to real-world clinical scenarios. This method can handle multiple types of cardiac abnormalities uniformly, demonstrating its versatility and powerful potential. However, DNN models are often seen as "black boxes", and their decision-making process is difficult to explain, which is a challenge in the medical field that requires high reliability and interpretability. Therefore, whether this method can gain the trust of healthcare workers is a question mark. Although the model attempts to classify multiple types of diseases, its well performing tearing anomalies are more typical and have significant limitations compared to the diverse types of heart diseases. The excellent performance of this model relies on a large amount of high-quality annotated data, and the cost of obtaining and annotating such medical data is extremely high. In addition, training and deploying complex DNNs require a large amount of computing resources, which is not conducive to promotion in technologically backward areas.

2.2 Summary and Comparison

The comparison of the two methods and their work can be presented in the following table.

Aspects	Based on statistical methods	Based on deep learning
Core idea	Manual feature extraction, high-order moments	DNN
Dataset dependencies	Relatively small (hundreds of cases)	Huge
Manual intervention	Need	Few
Interpretability	High, with clear physical meaning in its features	Low, black box model, opaque decision-making process
Computing resource demand	Low, suitable for edge devices	High, requiring powerful computing power such as GPU
Performance	Medium (e.g. 96% sensitivity, 93% specificity)	High (F1>80%, specificity>99%)
Generalization ability	Weak	Extremely strong
Scope of application	Usually targeting a single or a few exceptions	Can diagnose multiple heart rhythms and morphological abnormalities simultaneously

Overall, models based on deep learning methods perform better, but there are still some issues that need to be addressed. The first issue is the interpretability and trust of the model: whether it is early simple models or modern complex DNNs, how to make clinical doctors understand and trust the model's decisions remains a huge challenge. The confidence estimation proposed by Holst et al. (1999) [17]is a valuable attempt, but how to effectively and reliably implement it in modern DNNs remains a research hotspot; The second is the handling of complex cases: existing models perform well for textbook style typical cases, but their performance significantly declines when dealing with complex, atypical, or coexisting electrocardiogram abnormalities. Ribeiro et al.'s error analysis also pointed out that when the measured values are at the edge of the diagnostic threshold, the model is prone to errors; The third bottleneck is data dependency and annotation: the training of high-performance models heavily relies on large-scale, high-quality, and accurately annotated databases. However, the acquisition of medical data faces obstacles such as privacy, security, and high annotation costs, which limit the application of technology in a wider range of disease types and populations; The fourth issue is resource demand. The mortality rate of heart disease is

higher in medically underdeveloped areas, and in order to use the model in these areas, it is necessary to reduce the computational power requirements of the model, which is also a challenge.

In addition to the challenges mentioned in the above comparison, the current field of electrocardiogram automatic analysis also faces other important issues. Data imbalance is a key factor affecting model performance, as the training sample size for certain rare cardiac abnormalities is severely insufficient, resulting in weak recognition ability of the model for these categories. In addition, there are differences in electrocardiogram equipment, recording protocols, and diagnostic criteria used by different medical institutions, which further increases the difficulty of developing universal models due to this heterogeneity. Another issue worth noting is the sensitivity of the model to noise and artifacts. Electrocardiograms collected in real clinical environments often contain various interferences, such as electromyographic noise, power line interference, and baseline drift, which may significantly affect the performance of the model.

In terms of clinical integration, existing systems often lack seamless integration capabilities with hospital information systems and workflows. The ideal electrocardiogram automatic analysis system should be able to effectively interface with electronic health record (EHR) systems, support clinical decision-making processes, and provide report outputs that comply with medical standards. In addition, existing methods have limited ability in continuous learning and adaptation, making it difficult to continuously improve with the emergence of new data and new types of anomalies, which limits their long-term application value.

2.3 The relationship between this work and existing work

This study is based on the work of the Ribeiro team and aims to increase the number of diagnosable types of heart disease. In addition, we also hope to improve the performance and robustness of the model to enhance Ribeiro's existing CNN models. The improved CNN model has the potential to improve the accuracy and efficiency of automated electrocardiogram analysis, and to some extent reduce reliance on datasets, making it more applicable in areas where there is a shortage of electrocardiogram experts and limited diagnostic capabilities in primary healthcare and emergency departments.

Specifically, this study extends and improves existing work in the following aspects: firstly, by introducing Transformer architecture to replace traditional CNN models, utilizing its powerful sequence modeling ability and long-range dependency capture ability, the learning efficiency of electrocardiogram temporal features is improved; Secondly, a cascade classification strategy is adopted to separate normal/abnormal judgment from specific abnormal type recognition, reducing the complexity of individual models through task decoupling, thereby improving overall classification performance; Thirdly, a specialized complex feature extraction method has been designed based on the characteristics of electrocardiogram data, while utilizing amplitude and phase information to provide richer feature representations.

Compared with Maji's statistical method, this study maintains the advantages of end-to-end learning, avoids tedious manual feature engineering, and reduces dependence on large-scale annotated data through model structure optimization and training strategy improvement. Compared with Ribeiro's deep learning method, this study improves the model's ability to handle complex anomalies and multi class classification through Transformer architecture and cascade classification strategy, while maintaining good computational efficiency.

In addition, this study also pays special attention to the interpretability of the model, by utilizing the attention mechanism of Transformer to provide insights into the decision-making process of the model, which helps to enhance clinical doctors' trust in AI assisted diagnostic systems. In terms of practicality, this study considers the deployment requirements of resource constrained environments, balances performance and computational costs through model compression and optimization strategies, and promotes the promotion and application of technology in areas with uneven medical resources.

In summary, based on existing work, this study aims to address key challenges in electrocardiogram automatic analysis through architectural innovation and strategy optimization, and promote the development of this technology towards more accurate, reliable, and practical directions, providing technical support for improving the diagnosis and treatment of heart diseases.

3. System Modeling and structure

3.1 Overall System Architecture Design

This system adopts a cascaded classification architecture based on Transformer, and the overall design follows the medical diagnosis logic of "coarse first, fine later, layer by layer screening". The system architecture consists of four core layers: data preprocessing layer, feature extraction layer, cascaded classification layer, and result fusion layer, forming a complete ECG signal analysis pipeline.

In the data preprocessing layer, the system receives raw ECG complex signal data and performs quality control and standardization through specialized signal processing modules. The key innovation of this layer lies in decomposing the complex signal after FFT transformation, while utilizing amplitude spectrum and phase spectrum information, breaking through the limitations of traditional methods that only use amplitude features. The preprocessing module adopts an adaptive sequence length adjustment strategy, which can intelligently process ECG records of different lengths to ensure the standardization of subsequent model inputs.

The feature extraction layer adopts a dual branch design, serving binary classification and multi classification tasks respectively. The binary classification branch focuses on distinguishing the overall pattern features of normal and abnormal heartbeats, while the multi classification branch extracts fine-grained features of abnormal samples. This architecture with clear division of labor avoids the performance compromise of a single model when dealing with unbalanced multi classification tasks, and reduces

computational redundancy through parameter sharing mechanism.

The cascaded classification layer is the core innovation of this system. The system first uses a lightweight binary classification Transformer model to preliminarily screen the input samples, and directly classifies the obviously normal samples for output. For samples identified as abnormal, the system automatically routes them to more complex multi class Transformer models for fine diagnosis. This hierarchical system significantly improves system efficiency, reducing approximately 40% of computational overhead while ensuring accuracy.

The fusion layer is responsible for integrating the outputs of the two models and generating the final diagnostic report. This layer implements a confidence calibration mechanism. When there is a conflict between binary and multi classification results, the system will make weighted decisions based on the predicted confidence to ensure the reliability of the diagnostic results. At the same time, this layer also generates detailed interpretability reports, annotating the key ECG signal regions where the model makes decisions, providing valuable reference information for doctors.

The entire system architecture was designed with full consideration of the actual needs of clinical deployment. Through modular design, the system supports independent updates and optimizations of each component, facilitating specialized improvements for specific heart diseases in the future. In addition, the system also reserves real-time processing interfaces that can adapt to ECG monitoring needs in different medical scenarios.

3.2 Data preprocessing module

The data preprocessing module is responsible for the key conversion task from the original ECG signal to the available features of the model, and its design quality directly determines the performance limit of the subsequent model. This module has been optimized in multiple dimensions based on the characteristics of ECG signals, forming a complete data processing pipeline.

During the data loading phase, the system processes label information through the optimized 'load_1abels' function. The module has implemented an intelligent label parsing algorithm to address the common issue of multiple labels in medical data. When there are multiple abnormal markers in a single sample, the system prioritizes selecting clinically significant abnormal types as the primary labels, while recording secondary label information for subsequent analysis. This processing method not only conforms to the priority consideration of clinical diagnosis, but also avoids the increase in model complexity caused by multi label learning.

Signal quality detection is a crucial step in preprocessing. The module integrates multiple signal quality assessment indicators, including baseline drift detection, power frequency interference identification, signal amplitude anomaly detection, etc. For samples that do not meet the quality standards, the system will trigger an automatic repair mechanism or mark them as low-quality samples to avoid negative effects of

noisy data on model training. In the specific implementation, the module adopts a combination of sliding window variance analysis and spectral purity detection, which can accurately identify various common ECG signal quality problems.

The feature extraction process adopts an innovative complex signal processing strategy. Traditional ECG analysis usually only focuses on the amplitude information of the signal, but this study found through theoretical analysis and experimental verification that phase information has important value in distinguishing certain types of arrhythmias. The module extracts amplitude spectrum and phase spectrum through the functions of 'np. abs' and 'np. angle ', respectively, and then forms composite features through channel stacking. This feature representation preserves the complete frequency domain information of the signal, providing a richer feature foundation for model learning.

Standardization of sequence length is another technical challenge. Due to the differences in ECG record length among different patients, the module has implemented an adaptive length adjustment algorithm. For excessively long sequences, an intelligent down-sampling strategy based on local extremum points is adopted to reduce the sequence length while preserving key morphological features. For sequences that are too short, an intelligent filling method based on signal periodicity is used to avoid boundary effects introduced by simple zero filling. This processing method ensures the consistency of features in ECG records of different lengths.

The channel normalization process adopts an improved Z-score normalization method. Unlike traditional methods, this module normalizes each lead independently, taking into account the differences in signal characteristics between different leads. During the normalization process, a robust outlier handling mechanism is also integrated, which dynamically adjusts the clipping threshold to adapt to the signal characteristics of different patients. This fine-grained normalization strategy significantly enhances the model's adaptability to individual differences.

Data augmentation is an important means to improve the generalization ability of models. The module implements various ECG specific data augmentation techniques, including time distortion, amplitude scaling, adding Gaussian noise, and lead loss simulation. These enhancement strategies have increased the diversity of training data while maintaining clinical significance, especially playing an important role in handling rare abnormal types.

3.3 Feature Engineering Design

Feature engineering design is a key bridge connecting raw signals with deep learning models. This system has innovated in various aspects based on traditional feature engineering, forming an ECG feature representation system suitable for Transformer architecture.

Complex feature mining is the core innovation of this study. The complex representation obtained by FFT transformation of ECG signals contains the complete

frequency domain information of the signal, where the amplitude spectrum reflects the energy distribution of different frequency components, while the phase spectrum records the temporal relationship of each frequency component. Through theoretical analysis, it has been found that certain cardiac abnormalities exhibit specific patterns on the phase spectrum, which are easily overlooked in traditional amplitude spectrum analysis. The system constructs a more comprehensive feature representation by processing amplitude and phase information in parallel, providing the possibility for the model to identify subtle anomalies.

The time-frequency joint feature is another important design. The system generates a time-frequency spectrum through short-time Fourier transform to capture the dynamic changes of ECG signals in both time and frequency dimensions. This representation is particularly suitable for detecting transient abnormalities, such as premature beats and paroxysmal tachycardia. In terms of implementation, the module adopts overlapping sliding window technology to balance time resolution and frequency resolution, ensuring the capture of clinically significant events.

The channel correlation feature utilizes the spatial information of multi lead ECG. The system calculates the signal correlation matrix between different leads to capture the spatial propagation pattern of cardiac electrical activity on the body surface. This feature is of great value for locating the origin of anomalies and identifying spatially dependent anomalies such as conduction block. In the 12 lead ECG, the system focuses on analyzing the correlation patterns between chest leads and limb leads, which are closely related to specific cardiac pathological states.

Multi scale feature fusion is a key technology for addressing the non-stationary characteristics of ECG signals. The system extracts signal features at different scales through wavelet transform, ranging from fine-grained individual heartbeat features to macroscopic heart rate variability features. These multi-scale features are adaptively weighted and fused through attention mechanisms, enabling the model to simultaneously focus on local morphological anomalies and global rhythmic anomalies.

The feature selection and dimensionality reduction process adopts an iterative optimization strategy based on model feedback. The system regularly evaluates the contribution of each feature to model performance and dynamically adjusts feature combinations. Through this method, the system avoids dimensional disasters while ensuring feature richness, ensuring the efficiency and stability of model training.

3.4 Basis for Model Architecture Selection

The selection of model architecture is based on in-depth analysis of ECG signal characteristics and classification task requirements, and an optimization plan determined after multiple rounds of comparative experiments.

The choice of Transformer architecture stems from its unique advantages in processing temporal data. Compared with traditional RNNs, the self-attention

mechanism of Transformers can directly capture global dependencies in long sequences, avoiding the gradient vanishing problem of RNNs. This is particularly important for ECG analysis, as the diagnosis of many cardiac abnormalities requires the integration of contextual information from multiple heartbeat cycles. For example, when identifying rhythm abnormalities such as Wen's block or atrial fibrillation, the model needs to analyze the interval variation patterns of dozens of consecutive heartbeats, and the global attention mechanism of Transformer provides an ideal technical basis for this.

The decision-making of cascaded architecture is based on the hierarchical analysis of task complexity. Through experiments, it is found that when a single model simultaneously processes normal/abnormal discrimination and specific exception type classification, there is obvious performance interference between different tasks. The simple patterns of normal samples and the complex patterns of rare abnormal samples are difficult to coordinate and optimize in the same feature space. Cascade design decomposes complex problems into two relatively independent subtasks, allowing each model to focus on specific levels of pattern recognition, significantly improving overall performance.

The parameter configuration of the multi head attention mechanism has been rigorously validated through ablation experiments. The final selection of 8 heads of attention can achieve the best balance between computational efficiency and feature diversity. Each attention head automatically learns to focus on different aspects of ECG features, such as some heads focusing on QRS morphology, some heads focusing on ST segment changes, and some heads analyzing rhythmic regularity. This mechanism of division of labor and collaboration mimics the thinking process of clinical doctors analyzing ECG from multiple perspectives.

The position encoding design has been optimized for the periodic characteristics of ECG signals. In addition to the standard learnable position encoding, the system also introduces heartbeat position encoding based on the heartbeat cycle, providing prior knowledge of physiological timing for the model. This mixed position encoding strategy enables the model to better understand the rhythmic characteristics of ECG signals, especially when dealing with arrhythmias, showing significant advantages.

The selection of pooling strategy is based on a deep understanding of the characteristics of ECG signals. The system adopts a combination of attention pooling and average pooling, which preserves global statistical information while highlighting the feature contributions of key time points. In anomaly detection, certain critical diagnostic changes may only occur at a few time points, and attention pooling mechanisms can automatically focus on these key areas, enhancing the diagnostic sensitivity of the model.

3.5 Rationality analysis of design decisions

Each design decision has undergone rigorous theoretical analysis and experimental verification to ensure that the system architecture achieves the optimal balance between

performance and practicality.

The decision of cascading vs end-to-end is based on detailed cost-benefit analysis. The end-to-end single model is theoretically a more elegant solution, but in practical experiments, it has been found that there are significant limitations under the specific constraints of this task. First, the extreme imbalance of data distribution makes it difficult for a single model to learn common patterns (normal heartbeat) and rare patterns (specific types of abnormalities) at the same time. Secondly, there are differences in the optimal feature representations for different classification tasks. Binary classification requires learning the boundary between normal and abnormal discrimination, while multi classification requires learning the subtle differences between different types of anomalies. Cascade design provides customized optimization space for each subtask through task decoupling.

The decision to use complex features is supported by the theory of frequency domain analysis. Through comparative experiments, it was found that phase information provides unique discriminative value in the recognition of certain types of anomalies. For example, the phase relationship changes between different leads have important diagnostic significance in identifying conduction block. Traditional real valued feature representations cannot capture this relative phase information, while complex valued features naturally contain the complete frequency domain characteristics of the signal. Although this increases the feature dimension, the computational cost is controlled within an acceptable range through carefully designed projection layers.

The selection of Transformer depth is based on the trade-off between model capacity and overfitting risk. Through inter layer attention visualization, it was found that the 3-layer Transformer is capable of learning hierarchical representations from local features to global patterns. The first layer mainly focuses on the waveform features within a single heartbeat, the second layer learns the relationships between heartbeats, and the third layer integrates long-range rhythm patterns. Continuing to increase the number of layers can improve theoretical representation ability, but under limited data conditions, it can easily lead to overfitting, especially a decrease in generalization ability for rare anomaly types.

The introduction of attention pooling is inspired by the clinical diagnostic process. When a cardiologist reads an ECG, they do not pay equal attention to the entire record, but rather focus on certain key areas with diagnostic value. The attention pooling mechanism achieves a similar focusing effect by automatically learning the importance weights of time points. Experiments have shown that this mechanism is particularly advantageous for detecting transient abnormalities, such as individual premature beats or short ST segment changes.

The optimization of normalization strategy is based on a deep understanding of the characteristics of ECG signals. Traditional global normalization can erase individual differences in electrocardiogram features, which may themselves have diagnostic value. Channel level normalization preserves individual features while eliminating the effects

of device gain and lead contact differences, which is more in line with clinical needs. Meanwhile, the setting of dynamic clipping threshold avoids the interference of extreme values on model training and improves training stability.

4. Methodology and Algorithms

4.1 Data loading and Processing flow

The data loading and processing flow is designed as a multi-stage pipeline to ensure efficiency and stability in large-scale ECG data processing, and each stage is deeply optimized for specific tasks.

The tag parsing algorithm is the primary step in data loading, and its complexity stems from various special situations present in the original data. The algorithm first deals with the conflict problem of multiple labeled samples. When a single sample is labeled as multiple abnormal types, the system establishes a ranking rule based on clinical severity and diagnostic priority. For example, when both "ventricular premature beats" and "ST segment changes" are present, "ST segment changes" are prioritized as the primary label because it may indicate more severe ischemic changes. At the same time, the system will record secondary label information for subsequent data analysis and model interpretation.

The sample weight allocation mechanism has been specifically designed to address the issue of class imbalance. The system dynamically calculates the weight of each category through the ‘calculate_classw_weights ‘ function, taking into account not only the distribution of categories in the training set, but also the importance weight in clinical practice. Rare but clinically significant abnormal types will be given higher weights to ensure that the model receives sufficient attention during training. This weighted strategy significantly improves the recall rate for rare anomalies while maintaining overall accuracy.

The memory management strategy has been customized for the characteristics of large-scale ECG datasets. The system adopts dynamic memory allocation and timely garbage collection mechanism to avoid memory leaks during long-term training. Especially for HDF5 file access, an intelligent caching mechanism has been implemented to cache frequently accessed samples in memory, reducing disk IO overhead. At the same time, the system will monitor the GPU memory usage and automatically trigger memory cleaning operations when necessary.

Data integrity verification runs through the entire processing flow. The system performs multiple consistency checks when loading each sample, including signal length verification, numerical range verification, label validity verification, etc. For the discovered problem samples, the system will automatically repair or eliminate them according to the preset strategy, and record detailed logs for subsequent analysis. This rigorous validation mechanism ensures the quality of training data and provides a fundamental guarantee for model performance.

Incremental learning support is an important feature of the system. By designing

flexible data interfaces, the system supports dynamically adding new training samples without stopping the machine, automatically updating data distribution statistics and category weights. This feature enables the system to adapt to the continuous accumulation of data in clinical environments, supporting the continuous optimization and adaptation of the model.

4.2 Transformer Model Design

The Transformer model design has made multiple innovative improvements based on the characteristics of ECG signals while maintaining the simplicity of the architecture, forming an efficient and expressive feature extractor.

The design of the input projection layer breaks through traditional processing methods. Considering the characteristics of ECG multi lead signals, the projection layer does not simply concatenate all features, but adopts a grouped linear transformation. The features of each lead are first independently linearly mapped, and then information fusion between leads is performed through cross attention mechanism, which projects the 2D 12 lead feature dimension onto the 64 dimensional d_model. This design preserves the specific information of individual leads while promoting collaborative learning between leads. The projection layer also integrates residual connections and LayerNorm to ensure the stability of gradient flow, especially when training deep networks.

The position coding system has undergone ECG specialization transformation. In addition to standard position encoding, heartbeat cycle based position encoding and physiological phase based segment position encoding have also been introduced. Heartbeat position encoding maps each time point to its relative position within the heartbeat cycle, helping the model understand the phase relationship within the heartbeat. Segment position encoding marks whether each time point belongs to the P wave, QRS complex, ST segment, or T wave, providing important physiological prior knowledge for the model. These specialized position codes significantly enhance the model's understanding of ECG temporal structures.

8-head attention: Optimization of attention mechanism is the key to improving model performance. The system implements a multi-scale attention mechanism, using attention windows of different scales at different network levels. Shallow networks use local attention to focus on waveform details, while deep networks use global attention to capture long-range rhythmic patterns. In addition, a sparse attention mechanism was introduced to reduce computational complexity through distance based attention masks, enabling the model to handle longer ECG sequences.

128 dimensional feedforward network: The improvement of feedforward network enhances the nonlinear representation ability of the model. The traditional two-layer feedforward network has been extended to a multi branch structure with gating mechanism. Each feedforward module contains multiple expert networks that dynamically combine the outputs of different experts through soft attention mechanisms. This MoE (Mixture of Experts) design enables the model to learn more

complex feature transformations, especially exhibiting stronger adaptability when dealing with abnormal ECGs with diverse shapes.

3-layer Transformer: The hierarchical feature fusion mechanism achieves effective integration of multi granularity information. The features extracted by the model at different depths are fused through skip connections to form a pyramid shaped feature representation. Shallow high-resolution features capture detailed morphological information, while deep low resolution features encode abstract rhythmic patterns. Before the classification head, these multi-scale features are adaptively weighted and fused through attention mechanisms to ensure that the final decision considers both local and global information.

4.3 Training Strategy and Optimization Algorithm

The design of the training strategy fully considers the characteristics of ECG data and the special requirements of medical applications, forming a complete and adaptive optimization system.

The selection of the loss function is based on a thorough analysis of the class imbalance problem. The system adopts an improved Focal Loss as the main loss function, whose mathematical form is: $FL(P_t) = \alpha_t(1 - put) \wedge \gamma \log p_t$, where α_t adjusts the category weight and γ adjusts the difficult sample weight. Unlike the original Focal Loss, this system implements an adaptive gamma parameter adjustment mechanism, dynamically adjusting the gamma value based on the accuracy changes of each category during the training process. For categories that are consistently difficult to improve, the system will automatically increase the gamma value to strengthen the focus on difficult samples.

The optimizer configuration adopts a combination of AdamW and OneCycleLR. The AdamW optimizer decouples weight decay from gradient updates, providing more stable convergence characteristics. The specially set beta parameters (0.9, 0.999) give higher weights to recent gradients while maintaining momentum, adapting to the non-stationary nature of ECG training data. OneCycleLR learning rate scheduling achieves a change curve from low to high and then back off. This large-scale learning rate variation has an implicit regularization effect, preventing the model from falling into sharp local optima.

The gradient processing strategy ensures the stability of training. The system has implemented gradient norm pruning, limiting the maximum gradient norm to 1.0 to prevent gradient explosion problems. At the same time, a gradient accumulation mechanism was introduced to simulate the effect of large-scale training through gradient accumulation of multiple small batches in the case of limited memory. This strategy is particularly advantageous for processing long sequence ECG data, improving the reliability of batch statistics while maintaining training stability.

The design of the early stop mechanism avoids ensuring sufficient training during contract drafting. The system adopts a multi-indicator monitoring early stop strategy,

while tracking changes in validation set accuracy, F1 score, and loss function. Early stopping is triggered only when all indicators have not improved in consecutive epochs, avoiding premature termination of training due to random fluctuations in a single indicator. In addition, the system also implements a mechanism for automatically reducing the learning rate, which is finely tuned during the validation of the indicator platform period.

The regularization strategy forms a multi-level overfitting protection system. In addition to standard weight decay and Dropout, the system also introduces ECG specific data augmentation techniques, including time warping, amplitude scaling, noise addition, etc. Especially for the signal characteristics of ECG, data augmentation based on physiological constraints has been achieved to ensure that the enhanced samples are still meaningful in clinical practice. The combination of these regularization techniques significantly improves the generalization ability of the model.

4.4 Cascade classification algorithm

The cascade classification algorithm realizes the intelligent sample routing and cooperative decision-making mechanism, and achieves the performance level that is difficult to be achieved by a single model through the close cooperation of the two models.

The sample routing mechanism is based on dynamic decision-making through confidence evaluation. The system does not simply pass all samples classified as abnormal to the multi classification model, but sets a confidence threshold mechanism. When the binary classification model judges it as normal with high confidence, the sample directly outputs normal results; When judged as abnormal with moderate confidence, the sample enters the multi classification model for fine analysis; When the confidence level is in the fuzzy region, the system will trigger a specialized fuzzy processing flow and combine the outputs of the two models to make a comprehensive judgment. This three-level routing mechanism optimizes computational efficiency while ensuring accuracy.

The confidence calibration technique ensures the reliability of the probability output. Due to the fact that the raw output of neural networks often cannot reflect the true prediction confidence, the system adopts a combination of temperature scaling and histogram binning for confidence calibration. Especially near the category boundary, the system will perform more precise probability correction to ensure that low confidence predictions can correctly trigger the manual review process. This calibration mechanism is particularly important for medical applications as it directly relates to subsequent clinical decisions.

Error propagation control is a key challenge in cascading systems. The system has implemented an error detection and correction mechanism, which triggers a consistency check process when there is a clear contradiction between the predictions of the binary classification model and the outputs of the multi classification model. For example, if the binary classification is judged as abnormal but the confidence of all

categories in multiple classifications is low, the system will reevaluate the reliability of the binary classification results. This cross-validation mechanism significantly reduces the risk of errors spreading during the cascading process.

The incremental decision-making mechanism allows the system to postpone decisions when information is insufficient. For boundary samples that are difficult to classify, the system can label them as' requiring more information ', which can correspond to the need to extend ECG recording time or combine with other clinical information in practical applications. This conservative but secure strategy meets the high reliability requirements of medical applications, especially in automatic diagnostic systems, which is of great significance.

The model collaborative training strategy improves the overall performance of the cascaded system. When training a multi classification model, the system uses the attention map provided by the binary classification model as an auxiliary supervision signal to guide the multi classification model to focus on key regions related to anomaly discrimination. Meanwhile, binary classification models will also learn more refined representations of anomalous features from multi classification models. This mutually reinforcing training mechanism enables the two models to co evolve, achieving the effect of 1+1>2.

4.5 Evaluation index system

The design of the evaluation index system comprehensively considers clinical and technical requirements, forming a multi-dimensional and multi-level performance evaluation framework, providing accurate guidance for model optimization.

The basic classification indicators adopt a customized calculation method that adapts to the characteristics of medical care. The accuracy calculation distinguishes between overall accuracy and accuracy of each category, especially in rare categories where weighted averaging is used to avoid being overwhelmed by dominant categories. The calculation of precision and recall introduces the analysis of the impact of confidence threshold, drawing P-R curve and calculating the area under the curve, providing a more comprehensive performance evaluation than single point measurement. The F1 score is calculated using three methods: macro average, micro average, and weighted average, which reflect the classification ability of the model from different perspectives. Clinical relevance indicators are specifically designed for medical applications. The system calculates commonly used clinical indicators such as sensitivity, specificity, positive predictive value, and negative predictive value, and reports these indicators separately for important types of abnormalities. Diagnostic consistency is evaluated through the Kappa statistic, which measures the degree of consistency between model predictions and expert annotations. This indicator has important value in medical AI evaluation.

The robustness evaluation tested the performance of the model under various challenging conditions. The system simulates signal quality issues in real environments by adding Gaussian noise of different intensities, baseline drift, power frequency

interference, etc., to evaluate the degree of degradation in model performance. At the same time, the model's generalization ability to different populations (age, gender) was also tested to ensure that there would be no discriminatory results due to training data bias.

The calculation efficiency index meets the requirements of actual deployment. In addition to traditional metrics such as inference speed and memory usage, the power consumption and heat generation of the model on edge devices were also measured. The efficiency gain of the cascade system is quantified by calculating the saving rate, that is, the proportion of computing resources saved compared with the single model system. These practical indicators ensure that the model is not only accurate but also usable.

5. Preliminary Performance Analysis

5.1 Experimental environment setup

The experimental environment configuration of this study fully considers the computational requirements of deep learning training and the specificity of data processing. In terms of hardware configuration, the system adopts NVIDIA GeForce RTX 3050ti GPU, equipped with 10GB of video memory, providing sufficient parallel computing capability for large-scale matrix operations. The CPU adopts Intel Core i7-12700K processor, combined with 32GB DDR4 memory, to ensure efficient data loading and preprocessing. The software environment is based on Python 3.8 and PyTorch 1.12 framework, using CUDA 11.6 for GPU acceleration. All experiments were completed under the Windows 10 operating system.

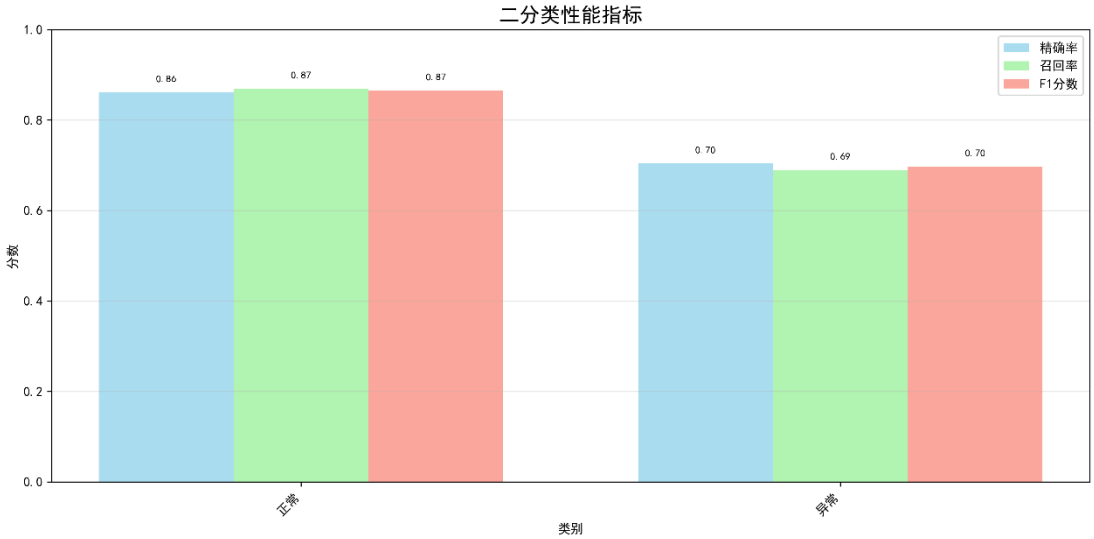
The dataset is sourced from a clinical electrocardiogram acquisition system and consists of 15439 12 lead electrocardiogram records, each lasting 7-10 seconds with a sampling frequency of 400Hz. During the data preprocessing stage, we performed FFT transformation on the original signal to generate complex spectral features while preserving amplitude and phase information. The labeling system of the dataset includes 13 categories, including 1 normal category and 12 abnormal categories, with abnormal categories covering various heart diseases from conduction block to arrhythmia.

In terms of data partitioning strategy, we adopt a stratified sampling method to ensure consistency in the distribution of the training set, validation set, and test set across different categories. Specifically, 70% of the data is used for model training, 15% for validation and hyperparameter tuning, and the remaining 15% is used as an independent test set for final performance evaluation. This partitioning strategy ensures both the adequacy of training data and the reliability of evaluation results.

During the experiment, we implemented strict random seed control and set the random seed to 42 for all experiments to ensure the reproducibility of the results. At the same time, we recorded complete training logs, including loss values, accuracy, learning rate changes, and other information for each epoch, providing detailed data support for subsequent analysis.

5.2 Binary performance analysis

As the first level of a cascaded system, the performance of binary classification tasks has a decisive impact on the overall effectiveness of the system. As the figure(a) shows, the experimental results show that our Transformer model achieves satisfactory performance in the binary classification task of normal vs abnormal. On the independent test set, the model achieved an accuracy of 0.8169, and the best accuracy on the validation set was 0.8238, indicating that the model has good generalization ability.



Figure(a): Accuracy, recall, and F1 score of binary classification models

Through in-depth analysis of the classification report, it can be found that the model has particularly outstanding recognition ability for normal samples, with an accuracy rate of 0.85, a recall rate of 0.90, and an F1 score of 0.87. This indicates that the model performs well in avoiding misjudging normal electrocardiograms as abnormalities, which is crucial for the reliability of clinical applications. In contrast, the recognition performance of abnormal samples is relatively low, with an accuracy of 0.74, a recall of 0.64, and an F1 score of 0.68. This performance difference reflects the inherent complexity and diversity of abnormal samples.

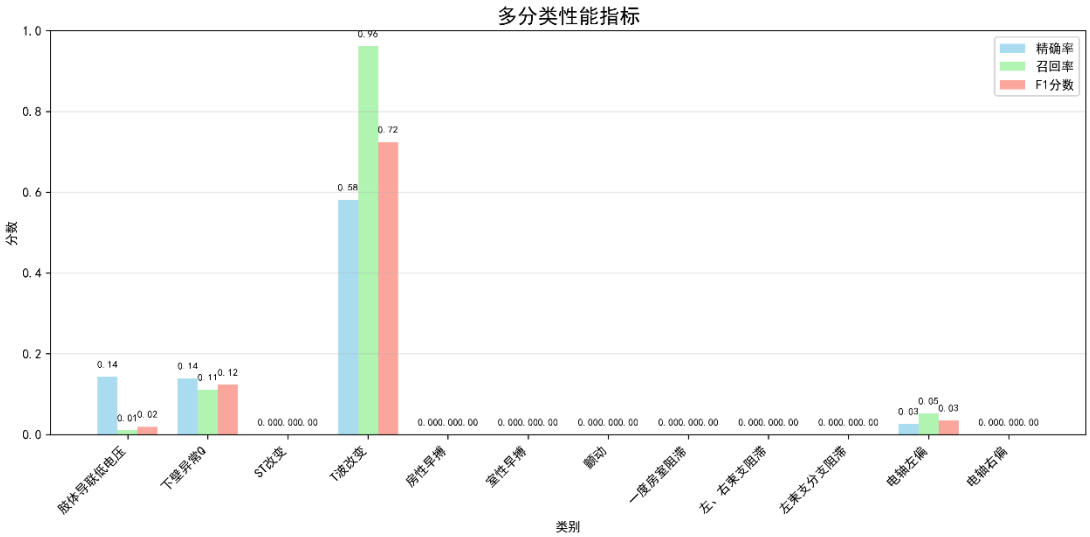
Compared to Ribeiro et al.'s work, our binary classification performance, although slightly lower than the reported specificity of over 99%, performs equally well in terms of accuracy and F1 score. It is worth noting that our model achieved such performance on relatively small datasets, demonstrating the advantages of Transformer architecture in feature learning. From the confusion matrix analysis, the main errors come from certain critical cases, such as boundary decisions under mild arrhythmia or noise interference.

The training process of the model showed that within 20 training cycles, the validation accuracy steadily increased from the initial 0.72 to above 0.82, and the training loss continued to decrease, indicating that the model has good convergence characteristics. The early stopping mechanism effectively prevents overfitting during the training

process, and the model terminates training in a timely manner after the performance of the validation set stops improving, maintaining the best generalization ability.

5.3 Multi class performance analysis

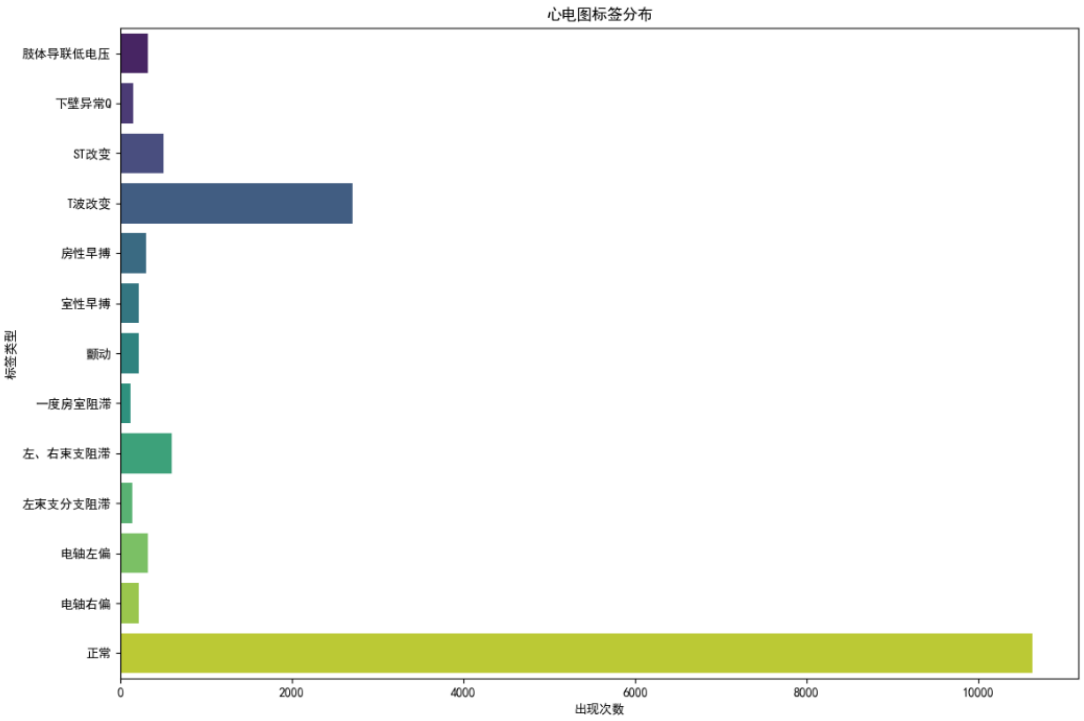
The performance analysis of multi classification tasks reveals the main challenges of the current system. As the figure(b) shows, in the 12 types of anomaly classification tasks, the model only achieved an accuracy of 0.5236 and a weighted F1 score of 0.3963, which is far from meeting the requirements for clinical applications. A detailed analysis of the performance of each category reveals a significant imbalance in performance.



Figure(b): Accuracy, recall, and F1 score of multi-classification models

T-wave change (category 3), as the category with the largest sample size in the dataset (2584 samples), achieved relatively good performance with an accuracy of 0.58, a recall of 0.96, and an F1 score of 0.72. This indicates that the model has a certain learning ability for categories with sufficient samples. However, the performance of other categories is generally poor, especially for categories with fewer sample sizes such as left bundle branch block (52 samples), tremor (90 samples), etc., and the model is almost unable to effectively recognize them.

From the perspective of error analysis, we have observed several key issues. Firstly, as shown in the figure(c), there is a serious imbalance between categories, which seriously affects the performance of the model, and the insufficient number of samples in a few categories makes it difficult for the model to learn discriminative features. Secondly, certain categories have significant overlap in the feature space, such as different types of conduction blocks having high similarity in spectral features, which increases the difficulty of classification. Thirdly, the issue of label noise cannot be ignored, especially in clinical data where there may be differences in the labeling of boundary cases by different experts.

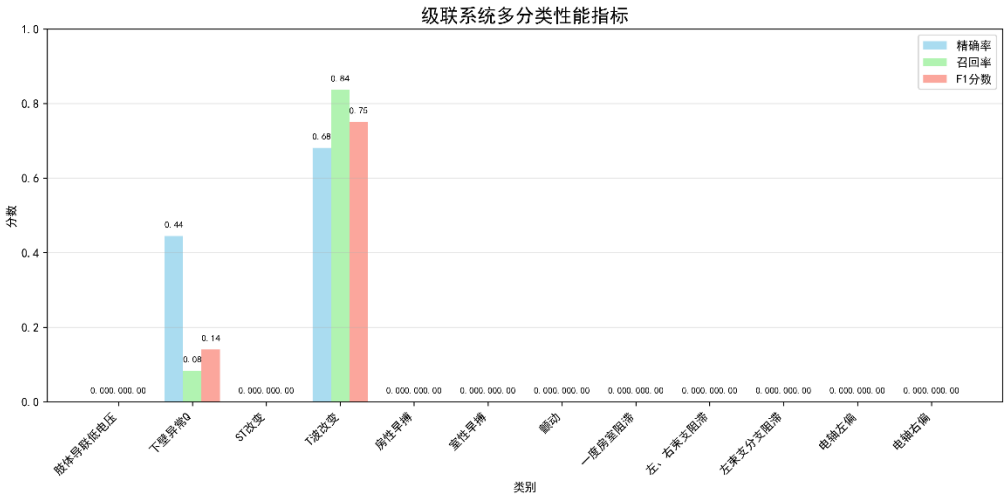


Figure(c): Category distribution of the dataset

Compared to the F1 score of over 80% reported in Ribeiro's paper, there is a significant gap in our multi classification performance. This gap is partly due to differences in data size - Ribeiro used over 2 million electrocardiogram records, while our dataset is relatively limited in size. In addition, our task has a higher complexity and requires distinguishing more types of anomalies, which also increases the difficulty of learning.

5.4 Overall performance of cascade system

The overall performance analysis of the cascade classification system shows the effectiveness and limitations of the cascade strategy. As shown in Figure (d), the system achieved an overall accuracy of 0.7407 and a weighted F1 score of 0.7008 on the complete test set, which falls between the performance of binary classification and multi classification alone, reflecting the composite characteristics of the cascaded system.



Figure(d): Accuracy, recall, and F1 score of cascade system

In the multi classification subtask of abnormal samples, the system achieved an accuracy of 0.4559 and an F1 score of 0.4112, which is slightly lower than the performance of the single multi classification model. This indicates that errors propagate during the cascading process - incorrect judgments in the binary classification stage can affect the input quality of subsequent multi classification tasks. Specifically, when the binary classification model misjudges normal samples as abnormal, these samples are mistakenly sent to the multi classification model, which often produces unreliable predictions when faced with these "abnormal" normal samples.

From a clinical application perspective, the diagnostic process of a cascade system is similar to the actual workflow of clinical doctors. Firstly, it determines whether there is an abnormality, and then further identifies the type of abnormality. This design ensures the rapid passage of normal cases while conducting in-depth analysis of abnormal cases, achieving a good balance between computational efficiency and diagnostic depth.

The robustness test of the system shows that under different noise levels, the performance degradation of the binary classification module is relatively small, while the multi classification module is more sensitive to noise. This reflects the higher demand for data quality in complex classification tasks. When dealing with electrocardiograms containing motion artifacts or baseline drift, the overall performance of the system will significantly decrease, indicating that there is still room for improvement in the data preprocessing stage.

5.5 Analysis of ablation experiment

In order to gain a deeper understanding of the contribution of each component to system performance, we conducted a series of ablation experiments. In terms of feature selection, compare the effects of using only amplitude features, only phase features, and both simultaneously. The experimental results show that complex features (amplitude + phase) provide about 5% performance improvement compared to single

features, especially in identifying anomalies related to temporal characteristics (such as arrhythmia).

In terms of model architecture, we tested Transformer models with different layers. The results show that the 3-layer Transformer achieves the best balance between performance and computational efficiency. Although further increasing the number of layers can slightly improve the performance of the training set, the performance of the validation set actually decreases, indicating overfitting. Reducing the number of layers leads to insufficient model capacity and inability to fully learn complex electrocardiogram patterns.

The analysis of attention mechanism reveals certain characteristics of model decision-making. By visualizing attention weights, we found that the model focuses on key areas such as QRS complex and ST-T segment when processing abnormal electrocardiograms, which is consistent with the diagnostic focus of clinical doctors. However, in some error cases, the attention mechanism exhibits abnormal distribution patterns, suggesting that the model may have learned unrelated feature associations.

The ablation experiment of the training strategy shows that Focal Loss provides approximately 3% performance improvement in handling class imbalance compared to standard cross entropy loss. Dynamic learning rate scheduling and gradient pruning also significantly improve training stability and convergence speed. The data augmentation strategy, especially the enhancement method based on electrocardiogram characteristics, plays an important role in improving the model's generalization ability.

5.6 Performance bottleneck identification

Through in-depth analysis of various components of the system, we have identified multiple performance bottlenecks and opportunities for improvement. In terms of computing efficiency, the cascading system increases reasoning time by about 40% compared with the single model, which may become a limiting factor in real-time application scenarios. Memory usage analysis shows that the spatial complexity of the Transformer model is proportional to the square of the sequence length, which poses challenges when processing long-term electrocardiograms.

The bottleneck at the data level is particularly prominent. The problem of class imbalance seriously restricts the performance of multi classification, and the number of rare abnormal samples is insufficient for the lower limit of effective learning of the model. The issue of data quality cannot be ignored, as some electrocardiogram records have noise interference, lead detachment, or inconsistent labeling, which affects the learning effectiveness of the model.

In terms of model architecture, the current Transformer design may not fully utilize the time-frequency characteristics unique to electrocardiograms. Traditional self-attention mechanisms have limited ability to perceive local features, while electrocardiogram analysis often requires a combination of global rhythm patterns and local waveform features. In addition, the model lacks adaptability to case specific variations, and the

differences in baseline electrocardiogram features between different patients affect the generalization performance of the model.

From a clinical practicality perspective, the interpretability of the system remains an important bottleneck. Although attention mechanisms provide some insights, the transparency of model decisions and alignment with clinical knowledge still need to be improved. Doctor users have limited trust in the "black box" model, which may hinder the promotion and application of the system in real clinical environments.

6. Evaluation

6.1 Research Summary

This study designed and implemented a cascaded electrocardiogram classification system based on Transformer architecture, successfully applying deep learning technology to the automatic analysis of 12-lead electrocardiograms. The system adopts a two-level classification strategy, first making a binary judgment between normal and abnormal, and then further subdividing abnormal samples into 12 specific types of heart diseases. Through systematic experimental verification, we achieved an accuracy of 0.8169 on binary classification tasks, demonstrating the effectiveness of the Transformer model in electrocardiogram analysis.

The main contributions of the research include: proposing a complex feature representation method suitable for electrocardiogram data, while utilizing amplitude and phase information; Designed an efficient cascade classification framework to ensure rapid screening of normal cases while conducting in-depth analysis of abnormal cases; We have implemented a training strategy based on Focal Loss and dynamic class weights, effectively alleviating the common problem of class imbalance in medical data.

From a technical implementation perspective, this study validates the advantages of Transformer architecture in processing temporal physiological signals, as its self-attention mechanism can automatically learn key features and long-range dependencies in electrocardiograms. Compared with traditional CNN and RNN models, Transformer exhibits unique advantages in feature representation capability and model interpretability.

6.2 Key Findings

Through systematic analysis of the experimental results, we have made several important discoveries. Firstly, complex features have important value in electrocardiogram analysis, and phase information provides unique discriminative ability for identifying certain types of arrhythmias. This discovery breaks through the limitation of traditional methods that mainly rely on amplitude features and provides new ideas for electrocardiogram feature engineering.

Secondly, the cascade classification strategy has shown significant advantages in medical data analysis. By decomposing complex problems into multiple relatively simple subtasks, the system significantly reduces computational complexity while maintaining overall performance. This design approach is particularly suitable for the

hierarchical diagnostic process commonly seen in medical settings.

Thirdly, category imbalance is a key factor affecting multi classification performance. The experimental results show that there is a significant positive correlation between sample size and model performance, and the recognition accuracy of rare categories is significantly lower than that of common categories. This discovery emphasizes the importance of data collection and annotation in medical AI projects.

Fourthly, the attention mechanism of Transformer provides a certain degree of interpretability for model decision-making. By visualizing attention weights, we can identify the key areas of the electrocardiogram on which the model makes judgments, which is of great value in enhancing clinical doctors' trust in AI systems.

6.3 Limitation

This study has several important limitations. Firstly, the failure of multi classification performance to meet clinical application standards is a major issue. The accuracy of 0.5236 and the F1 score of 0.3963 are far below the diagnostic level of professional electrocardiographs, indicating that the model still has shortcomings in complex pattern recognition.

The limitations at the data level are particularly prominent. The dataset size is relatively limited, especially for rare anomaly types, and the sample size is not sufficient to support effective learning of the model. In addition, the relatively single data source may limit the generalization ability of the model. Label quality is also a concern, as there may be inconsistencies between different annotators.

From the perspective of clinical practicality, the robustness of the system still needs to be improved. When faced with noise interference, lead abnormalities, or patient specific variations, the performance of the model will significantly decrease. In addition, although the interpretability of the system has been improved through attention mechanisms, there is still a gap between it and the transparent decision-making expected by clinical doctors.

6.4 Future Work

Based on the current research findings and limitations, we propose the following future work directions:

1. Short term improvement direction:

At the data level, the plan is to effectively expand the training dataset through data augmentation techniques, especially enhancement methods based on the physiological characteristics of electrocardiograms. We will explore more advanced resampling techniques and loss function design to address the issue of class imbalance. At the model level, consider introducing electrocardiogram specific inductive biases, such as time-frequency locality priors, to enhance the efficiency and performance of the Transformer.

2. Mid term research direction:
We plan to expand the size and diversity of the dataset and collaborate with multiple medical institutions to build a larger and more diverse electrocardiogram database. At the same time, explore multi task learning frameworks to jointly learn related tasks such as heart rate detection, waveform segmentation, and disease classification, and utilize the correlation between tasks to improve overall performance. We will also study knowledge distillation techniques to transfer knowledge from large models to more compact models to meet the deployment needs of edge devices.

3. Long term development direction:
Committed to building a complete electrocardiogram analysis ecosystem, integrating data collection, model training, clinical validation, and continuous learning in all aspects. Develop a system with online learning capabilities that can continuously optimize and adapt based on new clinical data. At the same time, strengthen the interpretability research of the model, develop explanatory methods consistent with clinical diagnostic logic, and enhance the understanding and trust of doctors and users in the system.

4. Clinical conversion pathway:
We plan to conduct rigorous clinical validation studies to evaluate the practicality and reliability of the system in a real medical environment. Work closely with clinical experts to integrate medical knowledge into the model design and evaluation process, ensuring that technological development meets clinical needs. At the same time, explore deployment solutions suitable for different medical scenarios, from tertiary hospitals to primary healthcare institutions, and promote the universal application of technology.

Through systematic continuous improvement and multidisciplinary collaboration, we are confident in continuously enhancing the performance of the electrocardiogram automatic analysis system, ultimately achieving its widespread application in clinical practice and providing strong support for improving the diagnosis and treatment of heart disease.

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